12TH CHEMISTRY · CHAPTER 20 · FULL CHAPTER MCQS

Aromatic Hydrocarbons

106 topic-wise MCQs from the complete chapter with answers and reasons

INTRODUCTION AND 20.1 STRUCTURE OF BENZENE

1. The simplest aromatic hydrocarbon is:

(a) Toluene
(b) Naphthalene
(c) Benzene
(d) Phenol

Answer: (c) Benzene — Formula C_6H_6.

2. The molecular formula of benzene is:

(a) C_6H_{12}
(b) C_6H_{10}
(c) C_6H_{14}
(d) C_6H_6

Answer: (d) C_6H_6 — Introduction.

3. The word “aromatic” originally meant:

(a) Cyclic
(b) Sweet smelling
(c) Unsaturated
(d) Stable

Answer: (b) Sweet smelling — Benzene derivatives are fragrant.

4. Hydrocarbons containing one or more benzene rings are called:

(a) Arenes
(b) Alkenes
(c) Alkynes
(d) Cycloalkanes

Answer: (a) Arenes — Introduction.

5. Vanillin is used as:

(a) A solvent
(b) A fuel
(c) An ice cream flavour
(d) A dye

Answer: (c) An ice cream flavour — It is a sweet-smelling aromatic compound.

6. Camphor has a strong aroma but:

(a) Lacks a benzene ring
(b) Contains two benzene rings
(c) Is an arene
(d) Is odourless when pure

Answer: (a) Lacks a benzene ring — It is non-aromatic.

7. The structure of benzene was proposed by Kekule in:

(a) 1825
(b) 1865
(c) 1905
(d) 1845

Answer: (b) 1865 — August Kekule, a German professor.

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8. Kekule's structure of benzene has:

(a) Six double bonds (b) A linear chain
(c) Only single bonds (d) Alternate single and double bonds

Answer: (d) Alternate single and double bonds — A six-membered ring.

9. Each carbon atom in benzene is:

(a) sp^2 hybridized (b) sp hybridized
(c) sp^3 hybridized (d) Unhybridized

Answer: (a) sp^2 hybridized — Planar hexagon.

10. The bond angles in benzene are:

(a) 109.5° (b) 120°
(c) 180° (d) 90°

Answer: (b) 120° — Both C–C–C and H–C–C.

11. The C–C bond length in benzene is:

(a) 154 pm (b) 133 pm
(c) 139 pm (d) 120 pm

Answer: (c) 139 pm — Between single and double bond lengths.

12. The C–H bond length in benzene is:

(a) 139 pm (b) 120 pm
(c) 154 pm (d) 109 pm

Answer: (d) 109 pm — From X-ray studies.

13. The C–C single bond length is:

(a) 133 pm (b) 154 pm
(c) 139 pm (d) 120 pm

Answer: (b) 154 pm — Table 20.1.

14. The C=C double bond length is:

(a) 154 pm (b) 139 pm
(c) 133 pm (d) 109 pm

Answer: (c) 133 pm — Table 20.1.

15. The shape of the benzene molecule is:

(a) Planar regular hexagon (b) Tetrahedral
(c) Linear (d) Pyramidal

Answer: (a) Planar regular hexagon — X-ray studies.

16. The heat of hydrogenation of cyclohexene is:

(a) 208 kJ mol^{-1} (b) 360 kJ mol^{-1}
(c) 240 kJ mol^{-1} (d) 120 kJ mol^{-1}

Answer: (d) 120 kJ mol^{-1} — It forms cyclohexane.

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17. The expected heat of hydrogenation of 1,3,5-cyclohexatriene is:

(a) 360 kJ mol^{-1} (b) 240 kJ mol^{-1}
(c) 208 kJ mol^{-1} (d) 152 kJ mol^{-1}

Answer: (a) 360 kJ mol^{-1} — 3 × 120.

18. The actual heat of hydrogenation of benzene is:

(a) 360 kJ mol^{-1} (b) 208 kJ mol^{-1}
(c) 120 kJ mol^{-1} (d) 152 kJ mol^{-1}

Answer: (b) 208 kJ mol^{-1} — Much lower than expected.

19. The resonance energy of benzene is:

(a) 208 kJ mol^{-1} (b) 360 kJ mol^{-1}
(c) 120 kJ mol^{-1} (d) 152 kJ mol^{-1}

Answer: (d) 152 kJ mol^{-1} — 360 – 208.

20. Resonance energy is also called:

(a) Activation energy (b) Lattice energy
(c) Delocalisation energy (d) Ionisation energy

Answer: (c) Delocalisation energy — Section 20.1.1.

21. The common product of hydrogenation of benzene and cyclohexenes is:

(a) Hexane (b) Cyclohexane
(c) Cyclohexene (d) Hexene

Answer: (b) Cyclohexane — The reference point.

22. The stability of benzene is due to:

(a) Its ring strain (b) Hydrogen bonding
(c) Delocalisation of π electrons (d) Its high mass

Answer: (c) Delocalisation of π electrons — Over all six carbon atoms.

23. The number of σ bonds formed by each carbon in benzene is:

(a) Three (b) Two
(c) Four (d) One

Answer: (a) Three — Two C–C and one C–H.

24. The unhybridized 2p orbitals in benzene are:

(a) In the plane of the ring
(b) Absent
(c) At 120° to the ring
(d) Perpendicular to the plane of the ring

Answer: (d) Perpendicular to the plane of the ring — They overlap sideways.

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25. The number of delocalised π electrons in benzene is:

(a) Three (b) Twelve
(c) Six (d) Four

Answer: (c) Six — One from each carbon.

26. The π-electron cloud in benzene lies:

(a) In the plane of the ring only
(b) Above and below the plane of the ring
(c) Outside the molecule
(d) On the hydrogen atoms

Answer: (b) Above and below the plane of the ring — Section 20.1.2.

27. The circle in the benzene hexagon represents:

(a) Six hydrogen atoms (b) The σ framework
(c) Three double bonds (d) Delocalised π electrons

Answer: (d) Delocalised π electrons — Section 20.1.2.

28. The C–H σ bond in benzene is formed by overlap of:

(a) sp^2–s
(b) sp^2–sp^2
(c) sp^3–s
(d) p–p

Answer: (a) sp^2–s — Carbon hybrid orbital with hydrogen 1s.

20.2 NOMENCLATURE OF BENZENE DERIVED COMPOUNDS

29. Hydrocarbons with only one benzene ring are:

(a) Monocyclic aromatic (b) Polycyclic
(c) Alicyclic (d) Aliphatic

Answer: (a) Monocyclic aromatic — Definition.

30. The common name of methylbenzene is:

(a) Aniline (b) Toluene
(c) Anisole (d) Phenol

Answer: (b) Toluene — Accepted by IUPAC.

31. The systematic name of phenol is:

(a) Benzenamine (b) Benzenecarbaldehyde
(c) Benzenol (d) Methoxybenzene

Answer: (c) Benzenol — OH on the ring.

32. Anisole is:

(a) Methylbenzene (b) Benzenol
(c) Benzenamine (d) Methoxybenzene

Answer: (d) Methoxybenzene — $C_6H_5OCH_3$.

33. Aniline is:

(a) Benzenol (b) Nitrobenzene
(c) Benzenamine (d) Benzonitrile

Answer: (c) Benzenamine — $C_6H_5NH_2$.

34. The systematic name of benzaldehyde is:

(a) Benzenecarboxylic acid (b) Benzenecarbaldehyde
(c) Benzenol (d) Phenylethanone

Answer: (b) Benzenecarbaldehyde — CHO on the ring.

35. Benzoic acid is also named:

(a) Benzenecarboxylic acid (b) Benzenesulphonic acid
(c) Benzenecarbaldehyde (d) Benzenol

Answer: (a) Benzenecarboxylic acid — COOH on the ring.

36. In a monosubstituted benzene the substituent is always at:

(a) C-2 (b) C-6
(c) C-4 (d) C-1

Answer: (d) C-1 — So no number is needed.

37. Which group has the highest priority in naming disubstituted benzenes?

(a) –OH (b) –COOH
(c) –CHO (d) $-NH_2$

Answer: (b) –COOH — First in the priority order.

38. In the priority order, –OH comes before:

(a) $-NH_2$ (b) –CHO
(c) –CN (d) –COOH

Answer: (a) $-NH_2$ — –COOH, –CN, –CHO, $-COCH_3$, –OH, $-NH_2$, –OR, –R.

39. If two substituents have the same priority, C-1 is given to the one that is:

(a) Heavier (b) More electronegative
(c) First in alphabetical order (d) Larger

Answer: (c) First in alphabetical order — Rule 2.

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40. The compound with Br and Cl on adjacent ring carbons is named:

(a) 2-Bromo-1-chlorobenzene
(b) Bromochlorobenzene
(c) 1-Chloro-3-bromobenzene
(d) 1-Bromo-2-chlorobenzene

Answer: (d) 1-Bromo-2-chlorobenzene — Bromo comes before chloro.

41. The prefix ortho- means the positions:

(a) 1,3 (b) 1,4
(c) 1,2 (d) 1,5

Answer: (c) 1,2 — Adjacent carbons.

42. The prefix meta- means the positions:

(a) 1,2 (b) 1,3
(c) 1,4 (d) 1,6

Answer: (b) 1,3 — Did You Know? box.

43. The prefix para- means the positions:

(a) 1,4 (b) 1,2
(c) 1,3 (d) 1,5

Answer: (a) 1,4 — Opposite carbons.

44. Naphthalene has the formula:

(a) $C_{14}H_{10}$
(b) $C_{12}H_{10}$
(c) C_6H_6
(d) $C_{10}H_8$

Answer: (d) $C_{10}H_8$ — Two fused rings.

45. Anthracene has the formula:

(a) $C_{14}H_{10}$
(b) $C_{10}H_8$
(c) $C_{12}H_{10}$
(d) $C_{16}H_{10}$

Answer: (a) $C_{14}H_{10}$ — Three fused rings.

46. An isomer of anthracene among the polycyclic arenes is:

(a) Naphthalene (b) Phenanthrene
(c) Toluene (d) Biphenyl

Answer: (b) Phenanthrene — Both are $C_{14}H_{10}$.

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47. TNT is:

(a) 1,3,5-Trinitrobenzene (b) 2,4-Dinitrotoluene
(c) 2,4,6-Trinitrotoluene (d) Trinitrophenol

Answer: (c) 2,4,6-Trinitrotoluene — Quick Check 20.2 (v).

20.3 REACTIVITY AND ELECTROPHILIC SUBSTITUTION

48. Alkanes typically undergo:

(a) Electrophilic addition (b) Electrophilic substitution
(c) Free radical substitution (d) Nucleophilic addition

Answer: (c) Free radical substitution — In UV light.

49. Alkenes typically undergo:

(a) Electrophilic substitution (b) Nucleophilic substitution
(c) Free radical substitution (d) Electrophilic addition

Answer: (d) Electrophilic addition — Available π electrons.

50. Benzene typically undergoes:

(a) Electrophilic substitution (b) Electrophilic addition
(c) Nucleophilic substitution (d) Elimination

Answer: (a) Electrophilic substitution — Aromaticity is retained.

51. Alkanes are unreactive because they are:

(a) Polar with π bonds (b) Non-polar with σ bonds only
(c) Ionic (d) Unsaturated

Answer: (b) Non-polar with σ bonds only — Section 20.3.

52. Benzene does not prefer addition because addition would:

(a) Need a catalyst (b) Form ions
(c) Release hydrogen (d) Break the delocalisation

Answer: (d) Break the delocalisation — The stability would be lost.

53. The first step of electrophilic aromatic substitution is:

(a) Generation of the electrophile (b) Loss of a proton
(c) Formation of the product (d) Loss of the catalyst

Answer: (a) Generation of the electrophile — Usually in situ.

54. The intermediate in electrophilic aromatic substitution is the:

(a) Carbanion (b) Sigma complex
(c) Free radical (d) Acylium ion

Answer: (b) Sigma complex — Also called arenium ion.

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55. The number of π electrons in the sigma complex is:

(a) Six (b) Two
(c) Four (d) Five

Answer: (c) Four — Two are used for the new σ bond.

56. In the sigma complex the positive charge is spread over:

(a) Five carbons (b) Six carbons
(c) One carbon (d) Three hydrogens

Answer: (a) Five carbons — Section 20.3.1.

57. The carbon bonded to the electrophile in the sigma complex is:

(a) sp^2 hybridized (b) sp hybridized
(c) sp^3 hybridized (d) Unhybridized

Answer: (c) sp^3 hybridized — It carries H and E.

58. In the last step, the C–H bond breaks by:

(a) Homolytic cleavage (b) Heterolytic cleavage
(c) Oxidation (d) Hydrolysis

Answer: (b) Heterolytic cleavage — The electrons return to the ring.

59. The final step of electrophilic substitution restores:

(a) The electrophile (b) The sigma complex
(c) The double bond count to four (d) Aromaticity

Answer: (d) Aromaticity — Loss of a proton.

60. Which is used as a halogen carrier in the bromination of benzene?

(a) $FeBr_3$ (b) H_2SO_4
(c) Ni (d) $KMnO_4$

Answer: (a) $FeBr_3$ — A Lewis acid.

61. The electrophile in the bromination of benzene is:

(a) Br^- (b) Br^+
(c) Br_2 (d) Br•

Answer: (b) Br^+ — $Br_2 + AlBr_3 \rightarrow Br^+ + [AlBr_4]^-$.

62. The by-product in the halogenation of benzene is:

(a) Hydrogen gas (b) Water
(c) A hydrogen halide (d) A halide salt

Answer: (c) A hydrogen halide — e.g., HBr.

63. Lewis acids used in halogenation must be:

(a) Hydrated (b) Basic
(c) Dilute (d) Anhydrous

Answer: (d) Anhydrous — Shown on the equation arrow.

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64. Nitration of benzene uses:

(a) Dilute HNO_3 only
(b) Conc. H_2SO_4 only
(c) Conc. HNO_3 and conc. H_2SO_4
(d) HNO_3 and $AlCl_3$

Answer: (c) Conc. HNO_3 and conc. H_2SO_4 — Nitrating mixture.

65. The electrophile in nitration is:

(a) NO_3^-
(b) HSO_4^-
(c) NO_2^-
(d) NO_2^+

Answer: (d) NO_2^+ — Nitronium ion.

66. In nitration, sulphuric acid acts as:

(a) A reactant that is used up
(b) A catalyst
(c) A reducing agent
(d) A solvent only

Answer: (b) A catalyst — It is regenerated.

67. In the nitrating mixture, nitric acid acts as:

(a) A base
(b) An acid
(c) A catalyst
(d) A reducing agent

Answer: (a) A base — It accepts a proton from H_2SO_4.

68. The temperature used for the nitration of benzene is about:

(a) 150 °C
(b) 0 °C
(c) 50 °C
(d) 300 °C

Answer: (c) 50 °C — Higher temperatures give more dinitro product.

69. At higher temperature nitration of benzene also gives:

(a) 1,2-Dinitrobenzene
(b) 1,3-Dinitrobenzene
(c) 1,4-Dinitrobenzene
(d) Aniline

Answer: (b) 1,3-Dinitrobenzene — $-NO_2$ is meta directing.

70. The catalyst is regenerated in nitration by:

(a) $NO_2^+ + H_2O$
(b) $H_2O + SO_3$
(c) $HNO_3 + H^+$
(d) $HSO_4^- + H^+ \rightarrow H_2SO_4$

Answer: (d) $HSO_4^- + H^+ \rightarrow H_2SO_4$ — Last step.

71. An acyl group is:

(a) R–C=O (b) R–
(c) R–O– (d) R–COO–

Answer: (a) R–C=O — An alkyl group attached to a carbonyl group.

72. Benzene with chloroethane and $AlCl_3$ gives:

(a) Chlorobenzene (b) Ethylbenzene
(c) Phenylethanone (d) Toluene

Answer: (b) Ethylbenzene — Friedel-Crafts alkylation.

73. The electrophile in Friedel-Crafts alkylation is a:

(a) Carbanion (b) Acylium ion
(c) Free radical (d) Carbocation

Answer: (d) Carbocation — Formed from the alkyl halide and $AlCl_3$.

74. The electrophile in Friedel-Crafts acylation is:

(a) An acylium ion (b) A carbocation
(c) A chloride ion (d) A nitronium ion

Answer: (a) An acylium ion — RCO^+.

75. The product of Friedel-Crafts acylation is an:

(a) Alkylbenzene (b) Aryl halide
(c) Aryl ketone (d) Aromatic acid

Answer: (c) Aryl ketone — e.g., phenylethanone.

76. In Friedel-Crafts reactions $AlCl_3$ acts as a:

(a) Lewis acid (b) Lewis base
(c) Reducing agent (d) Nucleophile

Answer: (a) Lewis acid — Electron pair acceptor.

77. The by-product in Friedel-Crafts reactions is:

(a) Water (b) HCl
(c) Cl_2 (d) H_2

Answer: (b) HCl — The proton joins a chloride ion.

78. Benzene with ethanoyl chloride and $AlCl_3$ gives:

(a) Ethylbenzene (b) Chlorobenzene
(c) Benzoic acid (d) Phenylethanone

Answer: (d) Phenylethanone — Friedel-Crafts acylation.

79. The most stable carbocation is:

(a) Methyl (b) Primary
(c) Tertiary (d) Secondary

Answer: (c) Tertiary — It forms most easily in alkylation.

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20.4 ORIENTATION AND OTHER REACTIONS

80. Groups that make the ring more reactive than benzene are:

(a) Deactivating
(b) Activating
(c) Meta directing
(d) Leaving groups

Answer: (b) Activating — They donate electrons.

81. Activating groups direct the incoming electrophile to the:

(a) Meta position
(b) Same carbon
(c) Side chain
(d) Ortho and para positions

Answer: (d) Ortho and para positions — 1,2 and 1,4.

82. Which is an ortho-para directing group?

(a) –OH
(b) $-NO_2$
(c) –CN
(d) $-CO_2-$

Answer: (a) –OH — It donates electrons to the ring.

83. Which is a meta directing group?

(a) –OH
(b) –R
(c) $-NO_2$
(d) –OR

Answer: (c) $-NO_2$ — Electron withdrawing.

84. Meta directing groups are mostly:

(a) Electron withdrawing
(b) Electron donating
(c) Alkyl groups
(d) Neutral

Answer: (a) Electron withdrawing — They destabilize the arenium intermediate.

85. Between ortho and para products, there is a slight preference for:

(a) Ortho
(b) Meta
(c) Para
(d) None

Answer: (c) Para — Section 20.4.

86. Chlorination of methylbenzene with $AlCl_3$ gives:

(a) 3-Chloromethylbenzene only
(b) Chlorobenzene
(c) Benzyl chloride
(d) 2- and 4-Chloromethylbenzene

Answer: (d) 2- and 4-Chloromethylbenzene — Methyl is 2,4-directing.

87. Ring chlorination of methylbenzene is done in the:

(a) Presence of UV light
(b) Absence of UV light
(c) Presence of nickel
(d) Gas phase at 500 °C

Answer: (b) Absence of UV light — With $AlCl_3$ or iron at room temperature.

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88. The –OH group directs incoming groups to positions:

(a) 2, 4 and 6 (b) 3 and 5
(c) 3 only (d) 1 only

Answer: (a) 2, 4 and 6 — It is a ring activator.

89. Chlorination of nitrobenzene gives mainly:

(a) 2-Chloronitrobenzene (b) 4-Chloronitrobenzene
(c) 3-Chloronitrobenzene (d) Chlorobenzene

Answer: (c) 3-Chloronitrobenzene — Nitro is meta directing.

90. Benzene burns with a:

(a) Colourless flame (b) Green flame
(c) Blue flame (d) Smoky flame

Answer: (d) Smoky flame — A test for aromatic compounds.

91. Complete combustion of 2 moles of benzene needs how many moles of oxygen?

(a) 12 (b) 15
(c) 9 (d) 6

Answer: (b) 15 — $2C_6H_6 + 15O_2 \rightarrow 12CO_2 + 6H_2O$.

92. Side-chain oxidation of toluene gives:

(a) Benzoic acid (b) Benzaldehyde only
(c) Phenol (d) Cyclohexane

Answer: (a) Benzoic acid — CH_3 becomes COOH.

93. Which reagent is used for side-chain oxidation?

(a) Nickel and hydrogen (b) $AlCl_3$
(c) Alkaline $KMnO_4$ (d) Bromine water

Answer: (c) Alkaline $KMnO_4$ — Also acidic $K_2Cr_2O_7$ or dilute HNO_3.

94. Side-chain oxidation of ethylbenzene gives:

(a) Phenylethanoic acid (b) Phenol
(c) Phenylethanone (d) Benzoic acid

Answer: (d) Benzoic acid — Whatever the length, one –COOH results.

95. For side-chain oxidation the alkyl group must have at least one:

(a) Double bond (b) Benzylic hydrogen
(c) Methyl branch (d) Halogen

Answer: (b) Benzylic hydrogen — On the carbon attached to the ring.

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96. Which compound is NOT oxidized by $KMnO_4$?

(a) Toluene
(b) Ethylbenzene
(c) (1,1-Dimethylethyl)benzene
(d) (1-Methylethyl)benzene

Answer: (c) (1,1-Dimethylethyl)benzene — A tertiary carbon has no benzylic hydrogen.

97. During side-chain oxidation, the purple colour of $KMnO_4$ is:

(a) Deepened (b) Discharged
(c) Turned blue (d) Unchanged

Answer: (b) Discharged — Used as an identification test.

98. Complete hydrogenation of benzene gives:

(a) Cyclohexane (b) Cyclohexene
(c) Hexane (d) Cyclohexadiene

Answer: (a) Cyclohexane — 3 moles of hydrogen add.

99. The catalyst for hydrogenation of benzene is:

(a) $AlCl_3$
(b) Conc. H_2SO_4
(c) $FeBr_3$
(d) Finely divided nickel

Answer: (d) Finely divided nickel — About 150 °C and high pressure.

100. Hydrogenation of methylbenzene gives:

(a) Methylcyclohexane (b) Cyclohexane
(c) Heptane (d) Toluene oxide

Answer: (a) Methylcyclohexane — The ring is saturated.

101. Compared with alkenes, hydrogenation of benzene needs:

(a) A lower temperature (b) A higher pressure
(c) No catalyst (d) UV light

Answer: (b) A higher pressure — High activation barrier.

102. Cyclohexane is used as a:

(a) Fuel additive only (b) Catalyst
(c) Solvent (d) Dye

Answer: (c) Solvent — Also a precursor for other chemicals.

103. Benzene reacts with chlorine in UV light and heat to give:

(a) Chlorobenzene (b) Benzyl chloride
(c) Dichlorobenzene (d) Hexachlorocyclohexane

Answer: (d) Hexachlorocyclohexane — An addition reaction.

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104. The addition of chlorine to benzene needs:

(a) $AlCl_3$
(b) Nickel
(c) UV light and heat
(d) Conc. H_2SO_4

Answer: (c) UV light and heat — No catalyst.

105. How many moles of chlorine add to one mole of benzene?

(a) One (b) Three
(c) Six (d) Two

Answer: (b) Three — $C_6H_6 + 3Cl_2 \rightarrow C_6H_6Cl_6$.

106. Phenol reacts with electrophiles faster than benzene because the –OH group:

(a) Donates a lone pair to the ring
(b) Withdraws electrons
(c) Is a meta director
(d) Blocks the ring

Answer: (a) Donates a lone pair to the ring — The ring becomes electron-rich.

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